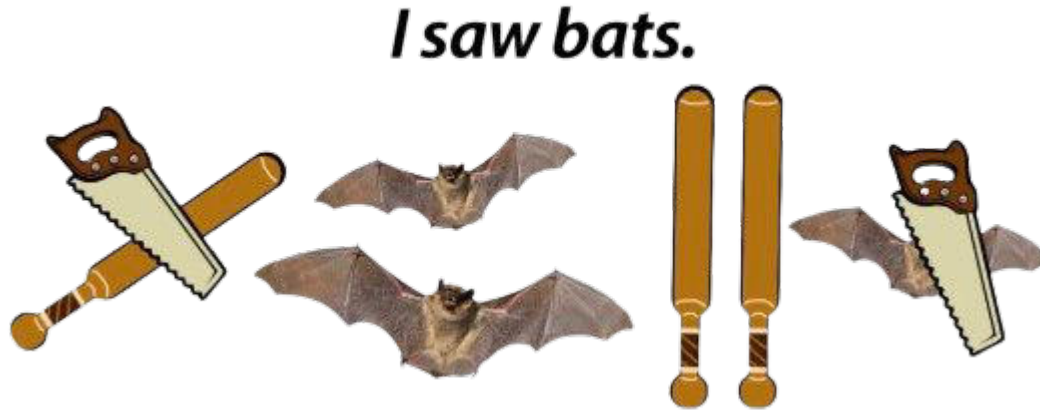


Ambiguity and Underspecification

Sampoorna Poria and Jihyung Park
November 4, 2025

What is Ambiguity?

- Ambiguity is the type of meaning in which a phrase, statement, or resolution is not explicitly defined, making for several interpretations



What is Underspecification?

- Underspecification is the lack of sufficient detail in a description, meaning, or representation.

Example -

1. When was *it* founded?
2. When did the *movie* come out?

The communicative function of ambiguity in language

Piantadosi et al., 2012

Why ambiguity isn't a flaw but a feature of efficient communication.

Ambiguity is pervasive

- Ambiguity appears at every level of language.
- Despite this, communication rarely fails.
- Suggests ambiguity is systematic, not accidental.

Why would languages evolve to keep ambiguity?



The Speaker–listener trade-off (Zipf's principle)

- Zipf's principle of least effort
- Ambiguity arises as a balance
- Context is informative

Information-theoretic explanation

- Ambiguity reduces redundancy
- Shannon entropy

$$\mathbf{H}[M] = - \sum_{m \in M} P(m) \log P(m)$$

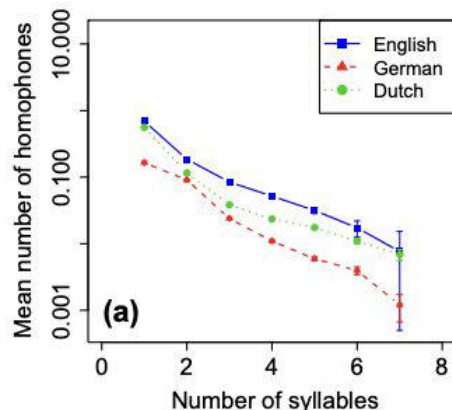
$$\mathbf{H}[M] > \mathbf{H}[M|C]$$

$$\mathbf{H}[M|C] = - \sum_{c \in C} P(c) \sum_{m \in M} P(m|c) \log P(m|c).$$

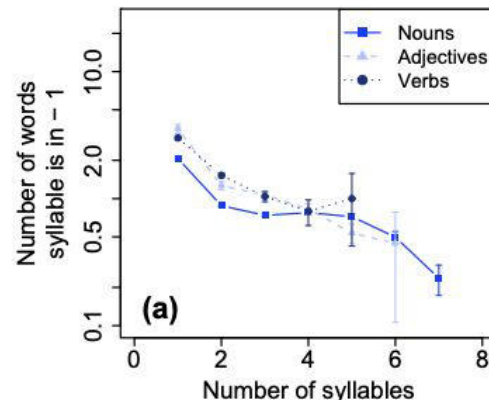
- Result: optimal communication uses ambiguous signals when context is highly informative, saving effort

Empirical Evidence - Form vs Ambiguity

- Observation in English/German/Dutch: “easy” words have more meanings.
- Short, frequent, phonotactically simple forms tend to be polysemous or homophonous.
- E.g. high-frequency words (like “run”) have many meanings, while long/rare words have fewer



Raw number of additional meanings each a phonological form has, as a function of length,



Raw number of additional senses a word has, as a function of a word's length,

Key takeaways

- Ambiguity is systematic and functional, not accidental.
- It results from clarity and efficiency trade-off
- Context enables this efficiency by resolving meanings.
- Predicts real-world patterns (word frequency $\leftrightarrow$ ambiguity).

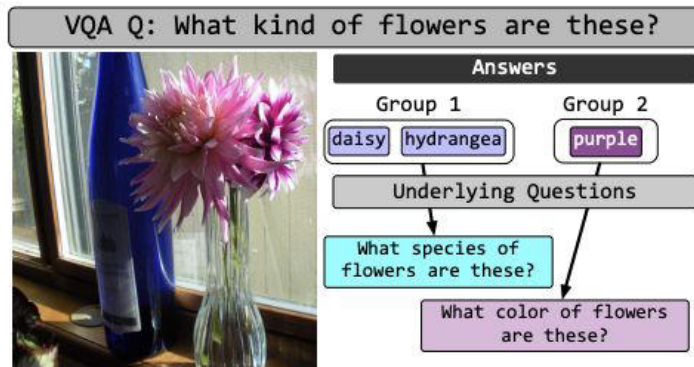
What happens when context is visual?

Why Did the Chicken Cross the Road? Rephrasing and Analyzing Ambiguous Questions in VQA

Stengel-Eskin et al., ACL 2023

Motivation & Examples

- Many visual questions (VQA) are linguistically ambiguous. Multiple valid questions fit the same phrasing.
- Ambiguity leads to annotator disagreements and QA errors.



Dataset Construction

- Subset of VQAv2 with ambiguous examples
- For each image-question, annotators group the multiple answers by the underlying question they imply.
- Then they then rewrite the question for each group so that it unambiguously matches that group's answers
- After cleaning examples, dataset has 629 unique image-question-rewritten question combinations (241 unique image-question combinations).

Types of Ambiguity (Ontology Overview)

Category	Property	PropB.	Description	Ex.
Property-based	Location	LOC	Asks about an object's location.	B.2.1
	Time	TMP	Asks about the time of an event or the time a picture was taken.	B.2.2
	Kind	N/A	Ask about what kind of something an object is.	B.2.3
Dynamic	Cause	CAU	Ask for the cause of an event.	B.3.1
	Purpose	PRP	Ask for the purpose of an event.	B.3.2
	Goal	GOL	Ask for the goal (location or person) of an object or event.	B.3.3
	Direction	DIR	Ask for the path being taken by an object.	B.3.3
	Manner	MNR	Ask in what manner an event is happening.	B.3.4
Pragmatic and Other	Multiple	N/A	Ask annotators to choose one of multiple options.	B.4.1
	Grouping	N/A	Ask annotators to group multiple items.	B.4.2
	Uncertainty	N/A	Contain visual uncertainty, especially for questions about events.	B.4.3
	Mistake	N/A	These involve bad answers or bad questions/images.	B.4.4

Question Generation Model

- Goal: Given an image and a (possibly ambiguous) question, generate a clarified question targeting a specific answer cluster.
- Architecture:
 - ViLT encoder
 - T5-base encoder-decoder
 - ViLT output replaces T5's embedding layer
- Training
 - on VQAv2 data (excluding annotated subset)

Underspecification

- Given an image and answer, there can be many possible questions that fit.
- Makes evaluation hard: different surface forms may all be semantically valid.
- The model is trained with the original question as the target (ground truth).
- Model needs cues to stay on topic while allowing rephrasing.

Solution: **Lexical Constraints**

- Extract noun phrases from the original question (e.g., bottle, cap, base).
- Generated question needs to include at least one of these noun phrases.



Answer: “purple”

Possible Questions

1. What color is the bottle?
2. What color is the bottle's base?

Performance

Model	BLEU-4	CIDEr	ROUGE-L	BERT
iVQA*	0.21	1.71	0.47	N/A
VT5-v	0.22	1.51	0.45	0.93
VT5-v+c	0.21	1.82	0.47	0.93
VT5-t	0.16	1.00	0.32	0.92
VT5-t+c	0.18	1.51	0.38	0.92
VT5	0.27	1.98	0.48	0.94
VT5+c	0.26	2.21	0.50	0.94

Table 2: Test performance of the VQG model and baselines. Our model is able to integrate multimodal information and produce high-similarity questions.



Performance

Method	Avg. P	Avg. R	Avg. F1
Human*	88.6	91.7	88.4
Random	64.9	70.4	59.4
Perfect P	100.0	50.6	61.1
Perfect R	63.4	100.0	76.3
GloVe initial	98.4	64.3	72.4
ViLT + K -means	65.9	68.6	60.1
VT5 + K -means	81.9	84.0	79.0

Table 3: Clustering metrics; Human results included for indirect comparison only.

Takeaways

- In VQA, lack of dialogue context amplifies ambiguity
- Understanding why questions are ambiguous helps design better annotation and generation frameworks.
- When evaluation accounts for ambiguity, models appear more competent and human-aligned.

Limitations / Discussion

- Subjectivity of annotations
- Quality of underlying data

What happens when context is textual and open-domain?

AMBIGQA: Answering Ambiguous Open-domain Questions

Min et al., EMNLP 2020

Motivation

- Real queries (e.g., Google search) often have multiple valid answers.
- Existing QA datasets (like NQ-OPEN) assume one true answer.
- Statistic: human agreement in NQ-OPEN test data = 49.2%.

Prompt Question

What would need to happen to change a lead atom into a gold atom?

Annotation #1

Question What more difficult process would need to happen to change a lead atom into a gold atom?

Answer nuclear transmutation | transmutation

Question What easier process would need to happen to change a lead atom into a gold atom?

Answer neutron capture and beta decay | neutron capture | beta decay

Wikipedia pages visited by annotators

[Nuclear transmutation](#) [Lead](#)

Original NQ answer

nuclear transmutation

Problem Definition (AmbigQA)

- Input: ambiguous question q
- Output
 - Set of all plausible answers $\{y_1 \dots y_n\}$
 - For each answer y_i , generate a disambiguated question x_i

Dataset Construction (AmbigNQ)

- Built from Natural Questions Open (NQ-Open).
- 14k questions; > 50 % ambiguous.
- Human annotators:
 - Found multiple valid answers in Wikipedia.
 - Wrote clarified questions for each.

Examples of Ambiguity Types

Type	Example
Event references (39%)	What season does meredith and derek get married in grey's anatomy? Q: In what season do Meredith and Derek get informally married in Grey's Anatomy? / A: Season 5 Q: In what season do Meredith and Derek get legally married in Grey's Anatomy? / A: Season 7
Properties (27%)	How many episode in seven deadly sins season 2? Q: How many episodes were there in seven deadly sins season 2, not including the OVA episode? / A: 25 Q: How many episodes were there in seven deadly sins season 2, including the OVA episode? / A: 26
Entity references (23%)	How many sacks does clay matthews have in his career? Q: How many sacks does Clay Matthews Jr. have in his career? / A: 69.5 Q: How many sacks does Clay Matthews III have in his career? / A: 91.5
Answer types (16%)	Who sings the song what a beautiful name it is? Q: Which group sings the song what a beautiful name it is? / A: Hillsong Live Q: Who is the lead singer of the song what a beautiful name it is? / A: Brooke Ligertwood
Time-dependency (13%)	When does the new family guy season come out? Q: When does family guy season 16 come out? / A: October 1, 2017 Q: When does family guy season 15 come out? / A: September 25, 2016 Q: When does family guy season 14 come out? / A: September 27, 2015
Multiple sub-questions (3%)	Who was british pm and viceroy during quit india movement? Q: Who was british viceroy during quit India movement? / A: Victor Hope Q: Who was british pm during quit India movement? / A: Winston Churchill

Evaluation Metrics

- Model prediction with m question-answer pairs compared with a gold reference set with n question-answer pairs
- Correctness score based on string similarity function f valued in $[0, 1]$

$$c_i = \max_{1 \leq j \leq n} \mathbb{I}[y_i \in \bar{\mathcal{Y}}_j] f(x_i, \bar{x}_j).$$

- $F1_{ans}$, $F1_{BLEU}$, $F1_{EDIT-F1}$

Model Architecture

- Built on Dense Passage Retriever (Karpukhin et al., 2020)
- Extended with:
 - Set-based QA: seq2seq model predicts multiple answers
 - Question Disambiguation: rewrites questions for each answer
 - Co-training: extra supervision from NQ-OPEN

Task 2: Question Disambiguation

- BART model takes $(q, \text{answer } y_i) \rightarrow$ generates disambiguated question x_i .
- Co-training: models label extra ambiguous data from NQ.
- Helps bootstrap learning without full annotation.

Results

Model	F1 _{ans} (<i>all</i>)		F1 _{ans} (<i>multi</i>)		F1 _{BLEU}		F1 _{EDIT-F1}	
	dev	test	dev	test	dev	test	dev	test
DISAMBIG-FIRST	28.1	24.8	21.9	18.8	4.2	4.0	2.7	2.2
Thresholding + QD	37.1	32.3	28.4	24.8	13.4	11.3	6.6	5.5
SPANSEQGEN + QD	39.7	33.5	29.3	24.5	13.4	11.4	7.2	5.8
SPANSEQGEN [†] + QD	41.2	35.2	29.8	24.5	13.6	10.6	7.4	5.7
SPANSEQGEN [†] (Co-training) + QD	42.3	35.9	31.7	26.0	14.3	11.5	8.0	6.3

Table 3: Results on AMBIGNQ. The *multi* measure only considers examples with multiple question-answer pairs.

Discussion & Implications

- Ambiguity is inherent in open-domain QA
- AmbigQA reframes QA: instead of finding one answer, models must return sets of plausible answers + disambiguated rewrites.
- Co-training with partial supervision improves learning
- Opens future directions in modeling ambiguity, event/entity reasoning, and improving user-facing QA clarity.

Ambiguity and Underspecification

November 4th, 2025
Jihyung Park

Before we start

Why Did the Chicken Cross the Road? Rephrasing and Analyzing Ambiguous Questions in VQA

Elias Stengel-Eskin Jimena Guallar-Blasco Yi Zhou Benjamin Van Durme
Johns Hopkins University
{elias, jguallal, yzhou188, vandurme}@jhu.edu

- Visual questions can be ambiguous due to linguistic or referential uncertainty
- **Approach:** Grouped diverse human answers, then asked annotators to rephrase each question to clarify its intended meaning.

AMBIGQA: Answering Ambiguous Open-domain Questions

Sewon Min,^{1,2} Julian Michael¹, Hannaneh Hajishirzi,^{1,3} Luke Zettlemoyer^{1,2}
¹University of Washington ²Facebook AI Research ³Allen Institute for Artificial Intelligence
{sewon, julianjm, hannaneh, lsz}@cs.washington.edu

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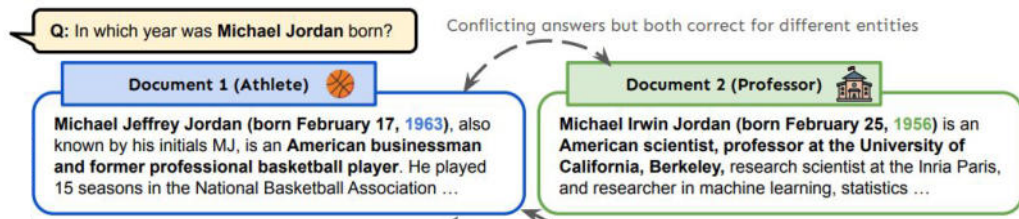
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But... is this enough for today's LLMs?

Ambiguity on RAG



Modern LLMs still hallucinate despite improvements.

RAG systems retrieve external knowledge but often:

- Misinterpret conflicting documents
- Fail to decide which one is correct

AmbigDocs: Reasoning across Documents on Different Entities under the Same Name

Yoonsang Lee^{◇,♡,*}, Xi Ye[◇], Eunsol Choi[◇]

The University of Texas at Austin[◇], Seoul National University[♡]

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COLM 2024

AmbigDocs: Background

Motivation: Ambiguity beyond the Question

Many entities share the same **surface name**.

When retrieving documents, **LLMs often mix information from distinct entities**.

Ambiguity is no longer only in the question.

Example:

“When is **ACL** 2024?”

-> One document: **Austin City Limits** Music Festival

-> Another: **Association for Computational Linguistics** Conference

AmbigDocs: Objective

AmbigDocs's question:

Can we **evaluate** whether **LLMs can reason correctly under document-level ambiguity**?

Key Idea:

When **multiple documents describe different entities** with the same name, can a model

1. **identify** which document corresponds to which entity?
2. **generate** distinct, factual answers for each?

Question: What is the state where **Judge Day** was born?

Doc 1 (**Charles Bernard Day**)
Charles Bernard Day is a ... Born in Dothan, **Alabama** ...

Doc 2 (**Edward William Day**)
Edward William Day was a ... Born in Cranston, **Rhode Island** ...

Doc 3 (**William Louis Day**)
William Louis Day was a ... Born in Canton, **Ohio** ...



LLM

AmbigDocs: Objective

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Dataset with ambiguous name and set of documents!

Wikipedia:Disambiguation

[Project page](#) [Talk](#)

From Wikipedia, the free encyclopedia

AmbigDocs: Dataset

Collect ambiguous surface names (S_n)

An **ambiguous entity name** that can be interpreted as any of disambiguated entities, depending on the context.

Retrieve Entity Documents

$\{d_1, d_2, \dots, d_m\}$

A **set of m documents** that are relevant to question q_{sn} that will be provided to the LMs.

For each entity listed, retrieve its Wikipedia page.

These pages from the **document set $\{d_1 \dots d_m\}$** for that question.

Each document corresponds to a distinct entity ($DE_1 \dots DE_n$).

Wikipedia Disambiguation Page

Hughesville

[Article](#) [Talk](#)

From Wikipedia, the free encyclopedia

Hughesville is the name of several places

- [Hughesville, Maryland](#)
- [Hughesville, Missouri](#)
- [Hughesville, New Jersey](#)
- [Hughesville, Pennsylvania](#)

AmbigDocs: Dataset

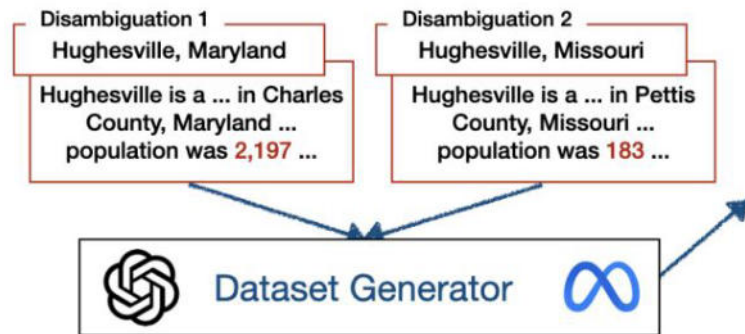
A list of disambiguated entities and Generate questions

A **set** of **n distinct entities** that share the same surface name **sn**.

Use **LLM prompting** to create **natural questions** containing the surface name.

“prompting it with (sn, d₁, d₂) along with two-shot exemplars to generate {q_{sn}, [a₁, a₂]}”

*“We initially use **GPT-4** (OpenAI, 2023), and then **distill** its ability to a smaller model, **Llama2-13B** (Touvron et al., 2023), by **finetuning with dataset generated from GPT-4** (West et al., 2021). We obtain a total of 43,839 questions after filtering low-quality examples.”*



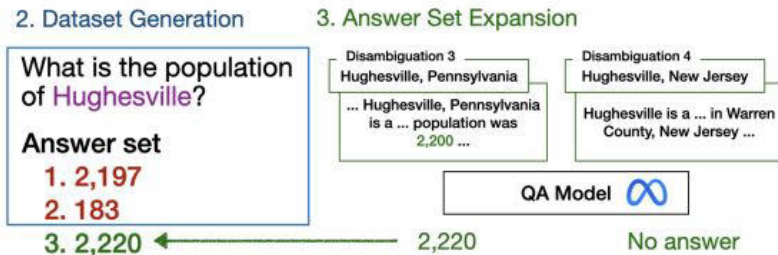
AmbigDocs: Dataset

Answer Set Expansion

The generated data contains only two answers belonging to two disambiguated entities while it can have more than two entities for each S_n .

As the QA module can generate an incorrect answer, they keep the newly **generated answer only when**

1. it **passes NLI entailment check**
2. **length-normalized log perplexity** of the answer is below a threshold value of **0.2**.



AmbigDocs: Evaluation

Answer for the given question using only the provided context.

Question: qsn

Context: $\{DE_{f(1)}^*\} \mid \{d_{f(1)}^*\}$

$\vdots$

Context: $\{DE_{f(m)}^*\} \mid \{d_{f(m)}^*\}$

Answer:

AmbigDocs: Evaluation

Answer Recall (AR)

How many **correct values** were mentioned

Entity Recall (ER)

How many **correct entity names** were recognized

$$\frac{1}{m} \sum R(a_i^*, y)$$

Entity-Answer Recall (EAR)

How many both **correct entity and correct answer** together

Disambiguation F1 (DF1)

How closely the **short answers extracted** by a RoBERTa QA model from an LLM's long-form response **match the gold answers**

$$DF1(y) = \frac{1}{m} \sum F1(a_i, a_i^*)$$

AmbigDocs: Results

	Llama2-7b	Mistral-7b	Llama2-13b	GPT-3.5	GPT-4
GOLD ONLY					
Answer Length ($ y $)	53.9	55.1	32.7	17.9	39.7
Answer Recall	0.536	<u>0.561</u>	0.441	0.393	0.632
Entity Recall	0.527	0.643	0.370	0.350	<u>0.596</u>
Entity-Answer Recall	0.372	<u>0.465</u>	0.244	0.234	0.473
Disambig-F1	0.219	<u>0.248</u>	0.178	0.180	0.289
GOLD+RETRIEVED					
Answer Length ($ y $)	67.0	59.9	36.0	18.4	40.6
Answer Recall	<u>0.495</u>	0.483	0.383	0.337	0.509
Entity Recall	<u>0.520</u>	0.571	0.351	0.319	0.482
Entity-Answer Recall	0.356	0.392	0.219	0.198	<u>0.358</u>
Disambig-F1	0.189	<u>0.199</u>	0.144	0.151	0.222
RETRIEVED ONLY					
Answer Length ($ y $)	61.4	58.8	33.2	18.2	37.7
Answer Recall	0.370	<u>0.384</u>	0.305	0.294	0.410
Entity Recall	<u>0.390</u>	0.454	0.289	0.281	0.385
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Even GPT4
achieved only
0.473 EAR!

AmbigDocs: Results

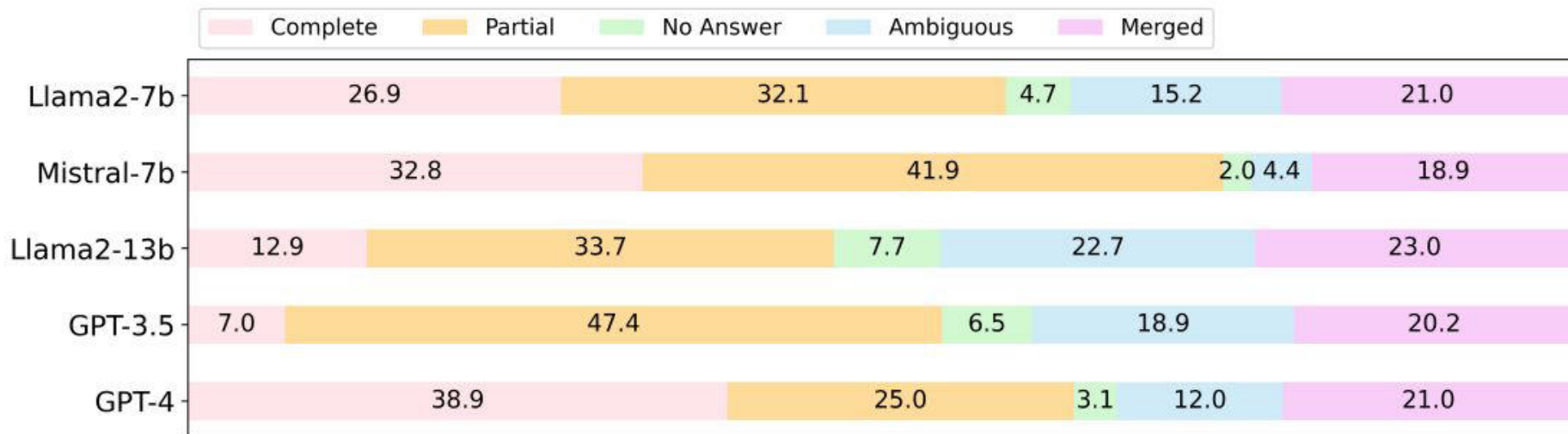
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Larger models are not always better perform!

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AmbigDocs: Results



AmbigDocs: Results

Example

q_{sn} : When did **Senator Grundy** serve in the United States Senate?

(DE_1^*, a_1^*) : (**Felix Grundy**, 1829 to 1838)

(DE_2^*, a_2^*) : (**Joseph R. Grundy**, December 11, 1929, to December 1, 1930)

Predictions

Llama2-7b: (*Merged*) Based on the provided context, **Senator Grundy** served in the United States Senate from **December 11, 1929, to December 1, 1930**, and again from **October 19, 1829, to July 4, 1838**.

Mistral-7b: (*Partial*) **Felix Grundy** served in the United States Senate from **October 19, 1829, to July 4, 1838**.

Llama2-13b: (*Ambiguous*) Based on the provided context, **Senator Grundy** served in the United States Senate from **October 19, 1829, to July 4, 1838**.

GPT-3.5: (*Ambiguous*) **Senator Grundy** served in the United States Senate from **October 19, 1829, to July 4, 1838**.

GPT-4: (*Complete*) **Joseph R. Grundy** served in the United States Senate from **December 11, 1929, to December 1, 1930**. **Felix Grundy** served in the United States Senate from **October 19, 1829 to July 4, 1838**.

AmbigDocs: Results - few shot

	Llama2-7b	Mistral-7b	Llama2-13b	GPT-3.5	GPT-4
Answer Recall	0.455 (-.081)	0.596 (+.035)	0.466 (+.025)	0.408 (+.025)	0.726 (+.094)
Entity Recall	0.444 (-.083)	0.723 (+.080)	0.442 (+.072)	0.397 (+.047)	0.796 (+.200)
Entity-Answer Recall	0.291 (-.081)	0.504 (+.039)	0.288 (+.044)	0.273 (+.039)	0.647 (+.174)
Disambig-F1	0.212 (-.007)	0.297 (+.049)	0.215 (+.037)	0.202 (+.022)	0.382 (+.093)

AmbigDocs: Summary

A benchmark to evaluate shows that **document-level ambiguity remains challenging for LLMs**

Result Summary:

Bigger models != Better

Llama2-13B, GPT-3.5 < Llama2-7B, Mistral-7B

Retrieval noise harms performance

GOLD + RETRIEVED < GOLD ONLY

In-context examples help

Two-shot improves Complete answers, largest gains in GPT-4

ZERO AND FEW-SHOT SEMANTIC PARSING WITH AMBIGUOUS INPUTS

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AMP: Background

Semantic parsing aims to map **natural language (NL)** into **logical forms (LF)** that represent meaning precisely, but **existing datasets (e.g., COGS, Spider) assume one unique LF per sentence.**

- Natural language is inherently ambiguous. **One sentence can have multiple valid interpretations.**
- Despite the importance of ambiguity in human language understanding, current LLMs and benchmarks evaluate models only on single “correct” mappings.
- As a result, how models handle ambiguous inputs remains underexplored.

AMP: Objective

AMP's question:

Can large language models **recognize and represent ambiguity** when translating natural language into logical forms without **explicit supervision**?

When given ambiguous examples in **few-shot prompts**, can models adjust their **output distribution to reflect the ambiguity ratio** in the data?

- Generate multiple valid logical forms (LFs) for an ambiguous sentence (Zero-shot setting)
- Adjust their prediction distribution when exposed to ambiguous examples in prompts (Few-shot setting)

AMP: Ambiguity Types

Type	Example Sentence	What is Ambiguous?
Prep. phrase attachment	The man saw the boy with the telescope.	Does “with the telescope” modify saw (instrument) or boy (possession)?
Quantifier scope	Every cow saw a dog	Does each cow see the same dog ($\forall x \exists y$) or different dogs ($\exists y \forall x$)?
Inverse Scope	A cow saw every dog.	Which quantifier takes wider scope — a or every?
Pronoun Coreference	Mary saw the woman and she smiled.	Does “she” refer to Mary or the woman?
Conjunction	The man drank and ate or swam.	Is it (drank and ate) or swam or drank and (ate or swam)?

AMP: METHODS - Dataset

- Constructed synthetic dataset of ambiguous sentences using **five linguistic templates**.
- Each template **generates two valid logical forms (LF_0 , LF_1)**.
- Lexical items (nouns, verbs, objects) are substituted from a fixed lexicon → **scalable generation**.
- Logical forms are represented in **First-Order Logic (FOL)** and **Lisp-style syntax**.

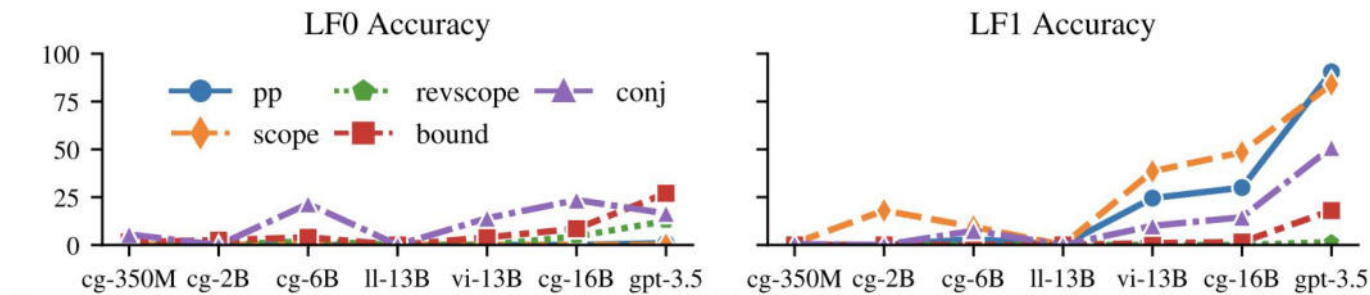
AMP: METHODS - Dataset

- **Zero-shot setting:**
 - Models receive no examples.
 - Goal: can the model **produce both valid LFs** (recognize ambiguity)?
 - Metric:
 - **ZM** – checks if LF_0 and LF_1 appear among top-k generations.

AMP: METHODS - Dataset

- **Few-shot setting:**
 - Prompts include mixed examples of LF_0 and LF_1 .
 - Goal: can the model **adjust its output distribution according to the ambiguity ratio?**
 - Metrics:
 - **FDM** (Few-shot Dataset Metric): how close the global LF ratio matches the prompt ratio.
 - **FIM** (Few-shot Instance Metric): how close each instance's LF_0 probability matches the prompt ratio.

AMP: Zero-Shot Result



Model	PP	Scope	Revscope	Bound	Conj.
cg-350M	0.00	0.00	0.00	0.00	1.00
cg-2B	0.00	0.00	0.00	0.50	0.00
cg-6B	0.00	0.00	0.00	0.00	3.50
cg-16B	0.00	0.00	0.00	3.50	15.00
ll-13B	0.00	0.00	0.00	0.00	0.00
vi-13B	0.00	0.00	0.00	4.00	9.50
gpt-3.5	0.00	0.00	0.00	0.00	0.00

AMP: Zero-Shot Result

Model	PP		Scope		Revscope		Bound		Conj.	
–	<i>FDM</i>	<i>FIM</i>	<i>FDM</i>	<i>FIM</i>	<i>FDM</i>	<i>FIM</i>	<i>FDM</i>	<i>FIM</i>	<i>FDM</i>	<i>FIM</i>
cg-350M	0.76	0.08	0.19	0.05	0.36	0.06	0.62	0.16	0.60	0.06
cg-2B	0.51	0.09	0.19	0.03	0.18	0.03	0.48	0.10	0.39	0.05
cg-6B	0.45	0.08	0.21	0.05	0.16	0.06	0.43	0.10	0.41	0.06
cg-16B	0.35	0.08	0.20	0.03	0.21	0.03	0.38	0.10	0.27	0.06
ll-13B	1.00	0.06	1.00	0.04	1.00	0.04	1.00	0.06	1.00	0.05
vi-13B	0.50	0.08	0.17	0.05	0.20	0.06	0.26	0.07	0.35	0.09

AMP: Summary

- LLMs **do not** inherently represent **ambiguity**.
- Even though models do not conceptually grasp ambiguity, they can still **reproduce** it if it exists in the data.

Limitations

- Synthetic nature
 - limited linguistic diversity (template-based).
- Not context-aware
 - single-sentence ambiguity only.

AMP: Own Questions

Bound pronouns For pronoun coreference, we show the transitive verb and each possible subject with the embedded verb separately. We also include an example of a subject with two verbs so that the model sees how to compose two verbs in the same sentence.

Let's translate what a human user says into what a computer might say.

Human: the woman saw Marie

Computer: exists x . exists a . woman(x) AND saw(a) AND agent(a, x) AND patient(a, Marie)

Human: Marie smiled

Computer: exists a . smiled(a) AND agent(a, Marie)

Human: the woman frowned and smiled

Computer: exists x . exists a . exists e . woman(x) AND frowned(a) AND agent(a, x) AND smiled(e) AND agent(e, x)

Human: the woman smiled

Computer: exists x . exists a . woman(x) AND smiled(a) AND agent(a, x)

Human: the woman saw Marie and she smiled

Computer:

Any Questions?